Juan Pablo de Legarreta

NETC 170 – IT Governance & Management

Module Ten Assignment

Business Impact Analysis Overview

Evaluating the business impact of a disaster requires an understanding of critical processes and systems for a business, assessing the impact of disruptions, and determining Recovery Time Objectives. (NineTech Consulting) These are critical to create an implementation plan for mitigation strategies in the event of a disaster. Each of the three scenarios will include a Business Impact Analysis that focuses on the impact of the situation on the business' critical processes and mitigation strategies for the future.

Scenario 1: Natural Disaster

The luxury resort that got hit by the hurricane has essentially lost all of its critical core processes. It cannot house guests given that many of its buildings have sustained damage and it has flooding and power outages. It also is facing huge financial strain due to lost revenue because of cancellations and high repair costs. By virtually eliminating its income sources and causing extensive damage, the hurricane has severely impacted the resort at every functional level. Fortunately, the resort was able to keep guests and employees safe throughout the disaster and coordinate with emergency services to ensure that its employees, at least, are safe.

To mitigate this risk in the future, the hotel can do the following:

1. Improve its physical infrastructure
    a. By structurally reinforcing the buildings on its property, the resort can ensure that a much more limited amount of damage will ensue from future storms.

b. By moving generators and electrical systems to a higher floor, it can minimize the impact of flooding.
c. The resort could invest in flood barriers or sea walls to help keep it safe from surge and flooding.

2. Secure its financial situation
    a. The resort could enhance its insurance to include things such as replacing assets and provide funds for business continuity.
    b. The resort could also create a financial plan to put a percentage of its revenue away as savings for an emergency fund.
    c. The hotel could start some sort of loyalty program whose income can go exclusively to disaster recovery money
3. Ensure that guests return more quickly
    a. In addition to strengthening the building and reconstruction, the hotel could create promotions and marketing to get guests to book in advance more quickly, asking for deposits or payment in full to boost its income

Scenario 2: Cyberattack

The investment bank experienced a thorough cyberattack that left customers unable to access their accounts or information and had their PII and financial information accessed by the hackers. Through the transparency of the bank's PR team, the customers were kept up to date with the status of the breach. The impact of the cyberattack was quite thorough – services went down, PII and financial information was exposed, and the bank also likely suffered a hit to its reputation that must have been rather public depending on how transparent the PR team really was. Since its core operations were unavailable to the public and the network was likely also inaccessible to the employees, the cyberattack almost completely ground the bank to a halt.

To mitigate this kind of attack in the future, the bank could:

1. Tighten Access Control for All Users
    a. The bank could add Multi-Factor Authorization for all users, requiring verification of identity to log into its systems.
    b. The bank could implement stricter role-based access policies, eliminating employees' access to only the resources they need to do their jobs.
    c. The bank could eliminate all global admin accounts but one and ensure that this account remains as little used as possible.
2. Improve Cybersecurity Tools and Training
    a. It's implied that the bank is investing heavily in cybersecurity measures and trainings; specifically, the bank could implement phishing training and phishing campaigns with its employees to try to reduce the likelihood of giving out credentials through phishing.
    b. The bank could also invest in a SIEM, especially one with AI, to try and give the security team a leg up on any events that are happening on the network.
    c. The bank could also redesign its network architecture and try to implement better network segmentation to create less possibility for lateral movement within a network.
3. Conduct a thorough Risk Assessment
    a. The bank could use a thorough Risk Assessment, especially for its third party risks. If the bank focuses on mitigating its third party risks, it will have a better

security posture for the future in case attacks come through its supply chain or third party vendors.

b. The bank could also get Cybersecurity insurance to try to curb its financial losses at a possible future attack and transfer some of the risk.

Scenario 3: Supply Chain Failure

The supply chain failure brought the auto manufacturer essentially to its knees. Not being able to source alternative components left the business with no alternative but to halt production and find a new alternative, which is a slow and expensive process, according to the case study. The halt in production also left its customers and dealers in a bind, damaging the company's relationship with them and its reputation. To mitigate this kind of situation again, the auto manufacturer could:

1. Obtain its electronic components from multiple sources
    a. The auto manufacturer can bid out contracts with multiple businesses and secure smaller contracts over several companies to ensure that its supply can be consistent and reliable even with fluctuations.
    b. The auto manufacturer can also search for companies in different parts of the world, so if economics or public policy change, it can pivot to its other sources without interrupting its supply or dramatically increasing its costs.
2. Create an Inventory Reserve Policy
    a. The auto manufacturer could create a policy of keeping a certain percentage of its stock in a rotating reserve in case of supply chain interruptions. This policy should come as a result of a risk assessment that gives a target amount of

inventory that could support a certain percentage of production for an acceptable time (e.g. have enough backups to operate at 65% capacity for 1 month)

   b. The auto manufacturer could use analytical tools to assess which components are most critical and ensure that it keeps an emergency supply of these components at all times.
   c. The auto manufacturer could move away from a strategy focused on ‘getting just as much as you need when you need it’ and instead go for a strategy of ‘having more than what we need before we need it’ without increasing its operating costs by too high a margin.

3. Invest in Risk Monitoring and Supply Management Tools
   a. The auto manufacturer could invest in supply chain management platforms that show analytics and real-time tracking to ensure that it can catch snags or problems before they occur.
   b. The auto manufacturer could also invest in risk monitoring services or tools that will help to use AI or predictive analysis to try and assess lead times, geopolitical instability, supply disruptions, or any other risk associated with its supply chain.

Conclusion

With the proper Risk Management strategies, each of these companies could mitigate disaster from hitting them as hard in the future. Utilizing a comprehensive business impact assessment focusing on the effects of disaster on its core functions and risk mitigation strategies, the companies can ensure that any future risk event can be handled with more ease and security. With the possibilities provided in this BIA, the organizations also have plenty of options to mitigate the risks that best fit their needs and budget.

Sources

NineTech Consulting. "How to Conduct a Business Impact Analysis (BIA): A Step by Step Guide." Linkedin.com, 27 May 2024, https://www.linkedin.com/pulse/how-conduct-business-impact-analysis-bia-step-by-step-84h6c/.